Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2019-II-Q02] | Part A (Q1-Q5)

A frog begins on pad one and independently jumps forward one or two pads with equal probability. If its probability of visiting pad seven is reduced p/q, find $p+q$.

青蛙从第一片荷叶开始，每次独立等可能前跳一片或两片。到过第七片的概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2023-I-Q06] | Part B (Q6-Q10)

A randomly shuffled deck contains three red and three black cards. Before each reveal, a player predicts its color using all previous reveals. Under an optimal strategy, the expected correct predictions equal the reduced fraction m/n. Find $m+n$.

三张红牌与三张黑牌随机洗匀后逐张翻开。每次翻牌前，可根据已翻牌预测颜色。最优策略的猜中次数期望为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2026-II-Q07] | Part B (Q6-Q10)

On each fair-die roll, outcomes $1, 2$ give a coin to Alice, $3, 4$ to Bob, and $5, 6$ to Carol. The chance that Alice and Bob both receive two coins before Carol receives her first is m/n in lowest terms. Find $100m + n$.

公平骰子每掷一次，$1, 2$ 给甲一枚硬币，$3, 4$ 给乙，$5, 6$ 给丙。甲乙均各得至少两枚而丙尚未得第一枚的概率为最简分数 m/n，求 $100m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2015-I-Q12] | Part C (Q11-Q15)

For every 1000-element subset of $\{1, \ldots, 2015\}$, record its smallest element. The mean of all recorded values is reduced p/q. Find $p + q$.

对 $\{1, \ldots, 2015\}$ 的每个 1000 元子集记录其最小元素。所有记录值的平均数为最简分数 p/q。求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-I-Q13] | Part C (Q11-Q15)

A lattice walker starts at height 21 between a reflecting fence $y = 0$ and absorbing river $y = 24$. Away from the fence it independently chooses east, west, north, south with equal probabilities. At the fence it chooses east, west, north equally. Find the expected number of unit jumps until it first reaches the river.

格点随机行走者从高度 21 出发，$y = 0$ 为不可越过的围栏，$y = 24$ 为到达即停止的河流。不在围栏上时，东、西、北、南四方向等可能；在围栏上时，东、西、北三方向等可能。求首次到河流所需单位跳跃次数的期望。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-II-Q13] | Part C (Q11-Q15)

Place six identical rooks on a 6×6 board with one in each row and column. An occupied cell in row r, column c has value $r + c$. The score is the minimum occupied value. If the mean score over all configurations is reduced p/q, find $p + q$.

在 6×6 棋盘放六个相同车，每行、每列各一个。第 r 行第 c 列格子的值为 $r + c$，一配置的得分为所占格子的最小值。所有配置平均得分为最简分数 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2018-II-Q13] | Part C (Q11-Q15)

Roll a fair six-sided die until the first occurrence of consecutive results $1, 2, 3$. If the probability the total number of rolls is odd is reduced m/n, find $m + n$.

反复投公平六面骰，首次连续出现一、二、三时停止。总投掷次数为奇数的概率为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2021-I-Q12] | Part C (Q11-Q15)

Three frogs start at vertices $4, 8, 12$ of a cycle of twelve vertices. Each minute each independently moves to either neighbor with equal probability. They stop at the first simultaneous landing that places two frogs together. If the expected stopping time is m/n reduced, find $m+n$.

十二顶点环上三只蛙初始位于第 $4, 8, 12$ 点。每分钟每只蛙独立等概率跳到左右相邻顶点之一，首次出现两蛙同时落在同一点时停止。若期望分钟数约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2025-I-Q13] | Part C (Q11-Q15)

Two perpendicular diameters divide a disk into four quadrants. Draw 25 independent random chords by choosing uniform perimeter endpoints conditioned on lying in different quadrants. Find the expected number of regions determined by all 27 segments.

两条垂直直径将圆盘分成四个象限。再独立画 25 条随机弦，每条的两端在圆周上均匀选取，条件是位于不同象限。求全部 27 条线段分出的区域数的期望。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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